

## **Appendix J: Assessed System Performance Reference**

This study report partially meets NERC TPL-001-4 standard, NERC FAC-002-2 standard, and ATC Planning Criteria (PLG-CR-0001-V20.2). In ATC's annual Ten-Year Assessment (TYA) and MISO annual MTEP studies, additional compliance related studies will be performed for the interconnection who will sign GIA.

Table J.1 – NERC FAC-002-2 Compliance Table

| Part | Item                                                         | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                     | Evidence Location                                                                                                                                                                                                                                                                                                                                                                      |
|------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1.2 | Reliability impact                                           | This study determines the reliability impact of 33 MISO generator interconnection requests (J1497 J1502 J1508 J1510 J1512 J1513 J1567 J1573 J1615 J1629 J1706 J1708 J1716 J1719 J1720 J1732 J1735 J1740 J1745 J1746 J1750 J1751 J1752 J1773 J1778 J1779 J1781 J1793 J1803 J1814 J1817 J1824 J1843 ) on the ATC System.                                                                                                                          | Section 2.3 covers steady state thermal and voltage analysis results which include thermal and voltage constraints impacted by the new generating facilities.<br>Section 3 and Section 6 covers dynamic/transient stability reliability impact of the new generating facilities<br>Section 4 cover short circuit and breaker duty reliability impact of the new generating facilities. |
|      | Applicable NERC Reliability Standards                        | The TPL-001-4 standard is selected as an applicable NERC Reliability Standard for this study. Portions of the reliability standard are applied when performing the system impact study for 33 MISO generator interconnection requests (J1497 J1502 J1508 J1510 J1512 J1513 J1567 J1573 J1615 J1629 J1706 J1708 J1716 J1719 J1720 J1732 J1735 J1740 J1745 J1746 J1750 J1751 J1752 J1773 J1778 J1779 J1781 J1793 J1803 J1814 J1817 J1824 J1843 ). | NERC TPL-001-4 Reliability Standard, as described in Appendix A Section A.1 and Appendix J Table J.2.                                                                                                                                                                                                                                                                                  |
|      | Regional planning criteria                                   | This study adheres to study requirements outlined in the applicable version of MISO BPM 015 – Generator Interconnection. BPM 015 contains MISO Business Practices in implementing Attachment X of the MISO Tariff. In addition, PJM Performs System Impact Study using their methodology.                                                                                                                                                       | MISO Business Practice Manual, BPM-015-r22, as described in Appendix A Section A.1                                                                                                                                                                                                                                                                                                     |
|      | Transmission Owner planning standards                        | This study takes into account the applicable version of the ATC Planning Criteria and Assessment Practices.                                                                                                                                                                                                                                                                                                                                     | ATC Transmission System Planning Criteria, PLG-CR-0001-V20.2, as described in Appendix A Section A.1<br>ATC Transmission Planning Assessment Practices, PLG-METH-0002-V20.1, as described in Appendix A Section A.1                                                                                                                                                                    |
| R1.3 | Facility interconnection requirements                        | This study also evaluates facility interconnection requirements such as voltage schedule, power factor, islanding detection and transformer energization in the applicable version of the ATC Generation Facility Interconnection Guide.                                                                                                                                                                                                        | ATC Generating Facility Interconnection Guide, Revision 11.0, as described in Appendix A Section A.1                                                                                                                                                                                                                                                                                   |
|      | Steady state analysis for normal and contingency conditions  | Steady State Power flow analysis determines impacts of new generating facility on the line flow and bus voltages. This study identifies violations to line ratings and bus voltage limits in study models under normal and contingency conditions.                                                                                                                                                                                              | The steady-state contingencies that were evaluated for the SIS are described in Appendix A. Section 2.3 covers steady state thermal and voltage analysis results.                                                                                                                                                                                                                      |
|      | Short circuit analysis for normal and contingency conditions | For existing substation and breakers, this study identifies areas where further investigation and potential upgrades and modifications will be required due new generator interconnections. It will be used to design new substation to be built for generator interconnections.                                                                                                                                                                | To be completed in later DPP phases.                                                                                                                                                                                                                                                                                                                                                   |
|      | Dynamic analysis for normal and contingency conditions       | This study simulates fault, relay and breaker actions near the generator and identifies any stability issues when new generators are interconnected to the grid. The results are used to develop mitigations and operating restrictions.                                                                                                                                                                                                        | To be completed in later DPP phases.                                                                                                                                                                                                                                                                                                                                                   |
| R1.4 | Study assumptions                                            | All relevant NERC Reliability Criteria and the ATC Planning Criteria are to be met for thermal, voltage and angular stability analysis.                                                                                                                                                                                                                                                                                                         | Appendix A covers study criteria, methodology and assumptions.<br>Section 2.1 covers modeling assumptions for the steady-state analysis.                                                                                                                                                                                                                                               |
|      | System performance                                           | See R1.3                                                                                                                                                                                                                                                                                                                                                                                                                                        | See R1.3                                                                                                                                                                                                                                                                                                                                                                               |
|      | Alternatives considered                                      | The consideration of alternatives was applicable to ERIS and NRIS network upgrade development and select point of interconnection facilities.                                                                                                                                                                                                                                                                                                   | Section 2.3.4 covers the alternatives that were considered for the ERIS thermal network upgrades.<br>Section 8.3 covers the alternatives that were considered for the NRIS thermal network upgrades.<br>Table J.3 covers the alternatives that were considered for point of interconnection facilities.                                                                                |
|      | Coordinated recommendations                                  | ATC will submit a generator interconnection project with a signed GIA, along with any required network upgrades to MISO. Then MISO will publish the generator interconnection project, along with any required network upgrades, in its MTEP Appendix A, which represents MISO accepts the projects and authorizes its implementation.                                                                                                          | To be completed in later MTEP process.                                                                                                                                                                                                                                                                                                                                                 |

Table J.2 – MISO, ATC, NERC/FERC Studies Matrix

|                               | MISO DPP |                 |     |     |     |     | ATC Planning Criteria |                    |     |     |                  |                  | NERC/FERC Compliance |                          |     |     |     |
|-------------------------------|----------|-----------------|-----|-----|-----|-----|-----------------------|--------------------|-----|-----|------------------|------------------|----------------------|--------------------------|-----|-----|-----|
| Study                         | PF       | DYN             | SC  | BD  | EMT | RAS | PF                    | DYN                | SC  | BD  | EMT              | RAS              | PF                   | DYN <sup>9</sup>         | SC  | BD  | RAS |
| P0                            | Yes      | Yes             | na  | na  | na  | na  | Yes                   | Yes                | na  | na  | na               | na               | Yes                  | Yes                      | na  | na  | na  |
| P1                            | Yes      | Yes             | na  | na  | na  | na  | Yes                   | Yes                | na  | na  | na               | na               | Yes                  | Yes                      | na  | na  | na  |
| P2                            | Yes      | Yes             | na  | na  | na  | na  | Yes                   | Yes                | na  | na  | na               | na               | Yes                  | Yes                      | na  | na  | na  |
| P3                            | NO       | NO              | na  | na  | na  | na  | Yes <sup>2</sup>      | Yes <sup>2</sup>   | na  | na  | na               | na               | Yes <sup>2</sup>     | Yes <sup>2</sup>         | na  | na  | na  |
| P4                            | Yes      | Yes             | na  | na  | na  | na  | Yes                   | Yes                | na  | na  | na               | na               | Yes                  | Yes                      | na  | na  | na  |
| P5                            | Yes      | Yes             | na  | na  | na  | na  | Yes                   | Yes                | na  | na  | na               | na               | Yes                  | Yes                      | na  | na  | na  |
| P6                            | NO       | NO              | na  | na  | na  | na  | Yes <sup>1</sup>      | Yes                | na  | na  | na               | na               | Yes <sup>1</sup>     | Yes                      | na  | na  | na  |
| P7                            | Yes      | Yes             | na  | na  | na  | na  | Yes                   | Yes                | na  | na  | na               | na               | Yes                  | Yes                      | na  | na  | na  |
| E1                            | NO       | NO              | na  | na  | na  | na  | Yes <sup>3</sup>      | Yes <sup>3</sup>   | na  | na  | na               | na               | Yes <sup>3</sup>     | Yes <sup>3</sup>         | na  | na  | na  |
| E2                            | NO       | NO              | na  | na  | na  | na  | Yes <sup>3</sup>      | Yes <sup>3</sup>   | na  | na  | na               | na               | Yes <sup>3</sup>     | Yes <sup>3</sup>         | na  | na  | na  |
| E3                            | NO       | NO              | na  | na  | na  | na  | Yes <sup>3</sup>      | Yes <sup>3</sup>   | na  | na  | na               | na               | Yes <sup>3</sup>     | Yes <sup>3</sup>         | na  | na  | na  |
| Power Factor Requirements     | Yes      | na              | na  | na  | na  | na  | Yes                   | na                 | na  | na  | na               | na               | Yes                  | na                       | na  | na  | na  |
| Transformer Energization      | na       | na              | na  | na  | na  | na  | Yes                   | na                 | na  | na  | na               | na               | na                   | na                       | na  | na  | na  |
| TPL-001-4 Requirement 4.3.1.3 | NO       | NO              | na  | na  | na  | na  | Yes <sup>2</sup>      | Yes <sup>2</sup>   | na  | na  | na               | na               | Yes <sup>2</sup>     | Yes <sup>2</sup>         | na  | na  | na  |
| Short Circuit                 | na       | na              | Yes | na  | na  | na  | na                    | na                 | Yes | na  | na               | na               | na                   | na                       | Yes | na  | na  |
| Breaker Duty                  | na       | na              |     | Yes | na  | na  | na                    | na                 | na  | Yes | na               | na               | na                   | na                       |     | Yes | na  |
| Models                        | SP&SH    | SH <sup>7</sup> | na  | na  | na  | na  | SP&SH                 | SP&LL <sup>2</sup> | na  | na  | na               | na               | SP&SH                | SP&LL <sup>2&amp;4</sup> | na  | na  | na  |
| Weak Grid Conditions          | na       | na              | na  | na  | NO  | na  | na                    | na                 | na  | na  | Yes <sup>5</sup> | na               | na                   | na                       | na  | na  | na  |
| SPS/RAS                       | NO       | NO              | na  | na  | na  | NO  | na                    | na                 | na  | na  | na               | Yes <sup>6</sup> | na                   | na                       | na  | na  | na  |
| Power Quality Analysis        | na       | na              | na  | na  | na  | na  | na                    | na                 | na  | na  | Yes <sup>8</sup> | na               | na                   | na                       | na  | na  | na  |
| LVRT (Order 661/661-A)        | na       | Yes             | na  | na  | na  | na  | na                    | Yes                | na  | na  | na               | na               | na                   | Yes                      | na  | na  | na  |

<sup>1</sup>ATC will perform NERC Category P6 steady-state thermal analysis to identify potential operating restrictions as part of DPP. ATC will perform NERC Category P6 steady-state voltage analysis outside of DPP process for the generators with signed the GIA in ATC Ten Year Assessment Process (TYA) annually.

<sup>2</sup>ATC will perform these studies outside of DPP process for the generators with signed the GIA in ATC TYA annually.

<sup>3</sup>ATC will rely on MISO annual MTEP Extreme Event study results for ATC TPL Extreme Event compliance. MISO MTEP study models include the generators with signed the GIA.

<sup>4</sup>TPL requires multiple models for dynamic analysis - ATC uses summer peak and light load models.

<sup>5</sup>ATC may perform electromagnetic transient (EMT) studies for select inverter-based resources in parallel with DPP studies.

<sup>6</sup>ATC will require assessment of existing SPS/RAS if there is potential for interaction with new interconnections.

<sup>7</sup>ATC will dispatch generators in the vicinity of new interconnections to Pmax.

<sup>8</sup>Developer is expected to provide power quality information to ATC for review. If deficient, studies will be required.

<sup>9</sup>Dynamic analysis may include EMT studies in selected situations.

Table J.3 – Point of Interconnection (POI) Alternatives Considered<sup>1</sup>

| MISO Queue # | POI Requested in Initial IC Application  | Alternative POI Considered                                                                                                                                                                                                                      | IC Final Decision on POI                 |
|--------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| J1497        | Concord 138 kV                           | N/A                                                                                                                                                                                                                                             | Concord 138 kV                           |
| J1502        | Briggs Road - North Madison 345 kV       | N/A                                                                                                                                                                                                                                             | Briggs Road - North Madison 345 kV       |
| J1508        | Wien - Stratford 115 kV                  | Stratford – McMillan 115 kV                                                                                                                                                                                                                     | Stratford – McMillan 115 kV              |
| J1510        | Arpin 345 kV                             | N/A                                                                                                                                                                                                                                             | Arpin 345 kV                             |
| J1512        | Darlington 138 kV                        | N/A                                                                                                                                                                                                                                             | Darlington 138 kV                        |
| J1513        | Highway 22 345 kV                        | N/A                                                                                                                                                                                                                                             | Highway 22 345 kV                        |
| J1567        | Werner West 138 kV                       | Werner 138 kV                                                                                                                                                                                                                                   | Werner West 138 kV                       |
| J1573        | Plover 115 kV                            | N/A                                                                                                                                                                                                                                             | Plover 115 kV                            |
| J1615        | Morgan 138 kV                            | N/A                                                                                                                                                                                                                                             | Morgan 138 kV                            |
| J1629        | Columbia - South Fond Du Lac 345 kV Line | Considered combining POI with J1705 (on same line), until J1705 withdrew,                                                                                                                                                                       | Columbia - South Fond Du Lac 345 kV Line |
| J1706        | North Monroe 138 kV                      | N/A                                                                                                                                                                                                                                             | North Monroe 138 kV                      |
| J1708        | J947 Interconnection Substation 138 kV   | N/A                                                                                                                                                                                                                                             | J947 Interconnection Substation 138 kV   |
| J1716        | South Fond Du Lac to Fitzgerald 345 kV   | Using J1253 (DPP-2019) Interconnection Substation                                                                                                                                                                                               | South Fond Du Lac to Fitzgerald 345 kV   |
| J1719        | Sand Lake 138 kV                         | N/A                                                                                                                                                                                                                                             | Sand Lake 138 kV                         |
| J1720        | Fountain Prairie -North Randolph 138 kV  | 138 kV line terminals (Academy, North Randolph) are beyond project site.                                                                                                                                                                        | Fountain Prairie -North Randolph 138 kV  |
| J1732        | ACEC Lewiston -Trienda 138 kV            | 138 kV line terminals (Trienda, Kilbourn) are beyond project ste.                                                                                                                                                                               | ACEC Lewiston -Trienda 138 kV            |
| J1735        | Paddock 138 kV                           | The IC expressed interest in a new substation on the Paddock – J1304 – Townline Rd 138 kV line.<br>ATC suggested utilizing the J1304 Interconnection Substation and the IC agreed<br>And submitted a new application reflecting the POI change. | J1304 Interconnection Substation 138 kV  |
| J1740        | University to Mukwonago 138 kV           | University-Whitewater 138 kV                                                                                                                                                                                                                    | University to Mukwonago 138 kV           |
| J1745        | Fitzgerald 138 kV                        | N/A                                                                                                                                                                                                                                             | Fitzgerald 138 kV                        |

|       |                                        |                                                                                                                                      |                                        |
|-------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| J1746 | Columbia 138 kV                        | N/A                                                                                                                                  | Columbia 138 kV                        |
| J1750 | Arnold – Dead River 345 kV             | Huron - Empire 138 kV                                                                                                                | Huron - Empire 138 kV                  |
| J1751 | ACEC Badger West - Saratoga 138 kV     | N/A                                                                                                                                  | ACEC Badger West - Saratoga 138 kV     |
| J1752 | Plover 115 kV                          | Golden Sands 138 kV                                                                                                                  | Golden Sands 138 kV                    |
| J1773 | Hill Valley to Cardinal 345 kV         | Hill Valley 345 kV Substation POI was suggested by ATC to the customer during the scoping call, but was not pursued by the customer. | Hill Valley to Cardinal 345 kV         |
| J1778 | Paris 138 kV                           | N/A                                                                                                                                  | Paris 138 kV                           |
| J1779 | Christiana – Kegonsa 138 kV            | ATC suggested Rockdale 345 kV to the customer and the customer accepted the POI change and updated their application.                | Rockdale 345 kV                        |
| J1781 | Hill Valley 345 kV                     | N/A                                                                                                                                  | Hill Valley 345 kV                     |
| J1793 | Holland 138 kV                         | N/A                                                                                                                                  | Holland 138 kV                         |
| J1803 | J1244 Interconnection Substation 69 kV | N/A                                                                                                                                  | J1244 Interconnection Substation 69 kV |
| J1814 | Nordic – Perch Lake 138 kV             | Nordic 138 kV                                                                                                                        | Nordic 138 kV                          |
| J1817 | J1244 Interconnection Substation 69 kV | N/A                                                                                                                                  | J1244 Interconnection Substation 69 kV |
| J1824 | Academy 138 kV                         | N/A                                                                                                                                  | Academy 138 kV                         |
| J1843 | Christiana 138 kV                      | N/A                                                                                                                                  | Christiana 138 kV                      |

<sup>1</sup> Based on MISO Tariff, when filing application to the interconnection queue, Interconnection customer (IC) can select any Point of Interconnection (POI) at a location under MISO’s functional control. This could be an existing substation or a new substation tapping the existing line. Further in the process, IC could unilaterally change POI to a new location within 5 days after the scoping call. Scoping calls occur within 45 days prior the start of Definitive Planning Phase (DPP). Any change in POI past the DPP start date requires consent for all parties i.e. MISO, TO, and IC.

Prior to the scoping calls, ATC review IC’s POI selection, initiates a conversation with IC and MISO and provides alternative POI options for consideration where warranted. ATC proposes alternative POI options for variety of reasons including, but not limited to, where the site constraints will make it difficult to expand an existing substation or build a new one, expanding the existing substation maybe be high cost compared to building a new site, connecting to a nearby existing substation maybe be an overall cheaper option compared to building a new site, or where combing several ICs relatively close to each other to a single site would be cost effective.

Ultimately, after the initial conversations, IC decides to pursue those option further or not. ATC offers help if further assistance is needed to evaluate various POI alternatives.